

Linux Fundamentals Part 1

TryHackMe Room Report
Cyber Security 101 – Linux Fundamentals

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Abstract

This report documents my work through the **Linux Fundamentals Part 1** room on TryHackMe, part of the *Linux Fundamentals* module. The room introduces the basics of the Linux operating system, including its history, deploying and connecting to a Linux machine, running fundamental commands, navigating the filesystem, and searching for files and text. As someone pursuing a path toward **penetration testing**, this room is especially relevant: Linux is the operating system underpinning the vast majority of pentesting tools, target servers, and distributions such as Kali Linux, making comfort with the command line a non-negotiable foundational skill.

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1 Introduction

Linux is a free, open-source operating system originally created by **Linus Torvalds** in **1991**. It is one of the most widely used operating systems in the world, powering everything from smart cars and Android devices to supercomputers and enterprise servers. Unlike Windows or macOS, Linux is highly customizable and comes in many different **distributions** (distros), such as Ubuntu, Fedora, Arch, and – most relevant to offensive security – **Kali Linux**, a distribution purpose-built for penetration testing.

Why This Matters for Pentesting

Almost every professional penetration testing toolkit (Kali Linux, Parrot OS) runs on Linux, and a huge proportion of real-world targets – web servers, cloud infrastructure, IoT devices – also run Linux. Mastering the command line here is not optional; it's the single most reused skill across every stage of a pentest engagement.

Answer

Research: What year was the first release of a Linux operating system?
1991

2 Task 3: Deploying a Linux Machine

TryHackMe provides an interactive, browser-based Linux virtual machine for this room, running **Ubuntu**. Deploying the machine provides:

- An assigned **IP address** for the machine.
- A **termination time**, after which the machine automatically shuts down.
- Direct access via an in-browser terminal (or optionally over SSH).

Note

No answer is required for this task – simply deploy the machine to follow along with the remaining exercises.

3 Task 4: Running Your First Few Commands

This task introduces two fundamental commands:

Command	Purpose
echo	Prints (outputs) text to the terminal.
whoami	Displays the username of the currently logged-in user.

3.1 Example Usage

```
tryhackme@linux1:~$ echo TryHackMe
TryHackMe

tryhackme@linux1:~$ whoami
tryhackme
```

Answer

If we wanted to output the text “TryHackMe”, what would our command be?

echo TryHackMe

What is the username of who you’re logged in as on your deployed Linux machine?

tryhackme

4 Task 5: Interacting with the Filesystem

This task covers navigating and inspecting the Linux filesystem using core commands:

Command	Purpose
<code>ls</code>	Lists the contents (files/folders) of the current directory.
<code>cd</code>	Changes the current working directory.
<code>pwd</code>	Prints the full path of the current working directory.
<code>cat</code>	Outputs the contents of a file to the terminal.

4.1 Example Usage

```
tryhackme@linux1:~$ ls
folder1 folder2 folder3 folder4

tryhackme@linux1:~$ cd folder4
tryhackme@linux1:~/folder4$ ls
todo.txt

tryhackme@linux1:~/folder4$ cat todo.txt
Hello World!

tryhackme@linux1:~/folder4$ pwd
/home/tryhackme/folder4
```

Why This Matters for Pentesting

Filesystem enumeration is one of the very first things a pentester does after gaining any level of access to a Linux target – looking for configuration files, credentials, SSH keys, and misconfigured permissions all starts with `ls`, `cd`, `cat`, and `pwd`.

Answer

On the Linux machine that you deploy, how many folders are there?

4

Which directory contains a file?

folder4

What are the contents of this file?

Hello World!

5 Task 6: Finding Files and Directories

This task introduces searching for content within the filesystem:

Command	Purpose
<code>find</code>	Searches for files/directories matching a given name or pattern.
<code>grep</code>	Searches for a specific string of text within a file.

5.1 Example Usage

```
tryhackme@linux1:~$ grep "THM" access.log
THM{FLAG_EXAMPLE}
```

Why This Matters for Pentesting

`grep` is one of the most-used tools during a real engagement – searching log files, source code, or configuration files for credentials, API keys, or sensitive strings (e.g. `grep -r "password"`) is a routine part of post-exploitation and enumeration.

Answer

Use the `cd` command to navigate to the file and find out the new current working directory.
What is the path?
/home/tryhackme/folder4
Use `grep` on `access.log` to find the flag that has a prefix of “THM”. What is the flag?
THMACCESS

6 Task 7: Shell Operators

This task introduces output redirection operators, which allow command output to be written to files instead of (or in addition to) the terminal.

Operator	Purpose
<code>></code>	Redirects output to a file, overwriting the file’s existing contents.
<code>>></code>	Redirects output to a file, appending to the existing contents.

6.1 Example Usage

```
# Overwrite the contents of "passwords" with "password123"
tryhackme@linux1:~$ echo password123 > passwords

# Append "tryhackme" to "passwords" while keeping existing content
tryhackme@linux1:~$ echo tryhackme >> passwords
```

Answer

If I wanted to replace the contents of a file named “passwords” with the word “password123”, what would my command be?
echo password123 > passwords
If I wanted to add “tryhackme” to this file named “passwords” but also keep “password123”, what would my command be?
echo tryhackme >> passwords

7 Task 8–9: Practical Challenge

The room concludes with a hands-on challenge that combines the commands learned throughout: creating a file, running a provided binary, and using the output to retrieve credentials for the next stage.

7.1 Approach

1. Use `touch` to create the required file (e.g. `noot.txt`).
2. Locate and execute the provided binary on the machine.
3. Read the binary's output to extract the required password/flag.

Why This Matters for Pentesting

This mirrors a real-world workflow: enumerate the filesystem, locate an executable or script, run it in a controlled way, and use its output to escalate further (e.g. move laterally, log in as another user, or access a restricted resource).

Answer

Terminate the machine deployed in this room once complete.
No answer needed – use the power button to terminate the machine.

8 Conclusion

Completing **Linux Fundamentals Part 1** provided a solid entry point into working with the Linux command line, covering:

- The history and background of Linux as an operating system.
- Deploying and connecting to a Linux virtual machine.
- Core commands: `echo`, `whoami`, `ls`, `cd`, `pwd`, `cat`.
- Searching the filesystem and file contents with `find` and `grep`.
- Redirecting command output using the `>` and `>>` shell operators.

Given my interest in **penetration testing**, this room is a critical building block: nearly every phase of a pentest – from initial enumeration to post-exploitation – relies on comfort navigating a Linux shell. I plan to continue directly into **Linux Fundamentals Part 2** and **Part 3** to build on filesystem permissions, package management, and more advanced shell operators, as part of my broader progression through the *Pre Security* and *Jr Penetration Tester* learning paths.

References

- TryHackMe – *Linux Fundamentals Part 1* room: <https://tryhackme.com/room/linuxfundamentalspart1>
- TryHackMe – *Cyber Security 101*: Linux Fundamentals module